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|-----------------------------------|--------------------------------------|
| <b>EXP NO 1</b>                   | <b>PRIVATE CLOUD USING OPENSTACK</b> |
| <b>Date:</b><br><b>24.06.2026</b> |                                      |

## **Aim**

To deploy and configure a single-node private cloud environment using OpenStack MicroStack and to study its major cloud services such as computing, networking, identity management, and storage. The experiment also involves creating and accessing virtual machine resources through the OpenStack web dashboard.

## **Introduction to OpenStack**

**OpenStack** is an open-source cloud computing platform used to build, deploy, and manage cloud infrastructure. It mainly provides **Infrastructure as a Service (IaaS)**.

OpenStack allows cloud administrators to manage various physical and virtual resources such as:

- Compute resources
- Storage resources
- Network resources

These resources can be allocated to users based on their requirements. Users can create and manage virtual machines and other cloud resources using the OpenStack platform.

OpenStack is commonly used for building **private and hybrid cloud environments**. In contrast, platforms such as AWS and Microsoft Azure mainly provide public cloud services.

## **What is MicroStack?**

**MicroStack** is a lightweight and simplified distribution of OpenStack that allows OpenStack to be installed and configured with minimal effort.

MicroStack is mainly useful for:

- Learning OpenStack concepts
- Testing cloud infrastructure
- Development and experimentation
- Running OpenStack on a single computer or virtual machine

In this experiment, MicroStack is deployed on **Ubuntu 22.04 LTS** running inside VMware. Since this is a single-node deployment, the required OpenStack services operate on the same system.

## Major OpenStack Services

| OpenStack Service | Main Purpose                                                                          |
|-------------------|---------------------------------------------------------------------------------------|
| Nova              | Creates, runs, and manages virtual machines                                           |
| Neutron           | Provides networking services such as virtual networks, IP addresses, and connectivity |
| Keystone          | Provides authentication, authorization, and identity management                       |
| Glance            | Stores and manages virtual machine images used for creating instances                 |
| Horizon           | Provides a web-based graphical dashboard for managing OpenStack resources             |
| Cinder            | Provides persistent block storage that can be attached to virtual machines            |
| Swift             | Provides object-based storage for storing and managing data                           |

## Software / Environment Used

- Operating System: Ubuntu 22.04 LTS (Jammy)
- Cloud Platform: OpenStack MicroStack
- Virtualization Platform: VMware
- Deployment Type: Single-node OpenStack
- Interface: Horizon Web Dashboard

## Procedure

### Step 1: Install MicroStack

Open the Ubuntu terminal and execute the following command:

```
sudo snap install microstack --beta
```

This command downloads and installs the MicroStack package along with the required OpenStack components.

### Step 2: Verify the Installation

After installing MicroStack, verify the installation using the following command:

```
snap list microstack
```

The command displays information about the installed MicroStack package and its version.

### Step 3: Install Python Compatibility Package

Install the required Python compatibility package using:

```
sudo apt install python-is-python3
```

This ensures that the python command refers to the installed Python 3 interpreter.

### Step 4: Initialize MicroStack

This command automatically configures the required OpenStack services and prepares the system for use as a single-node private cloud.

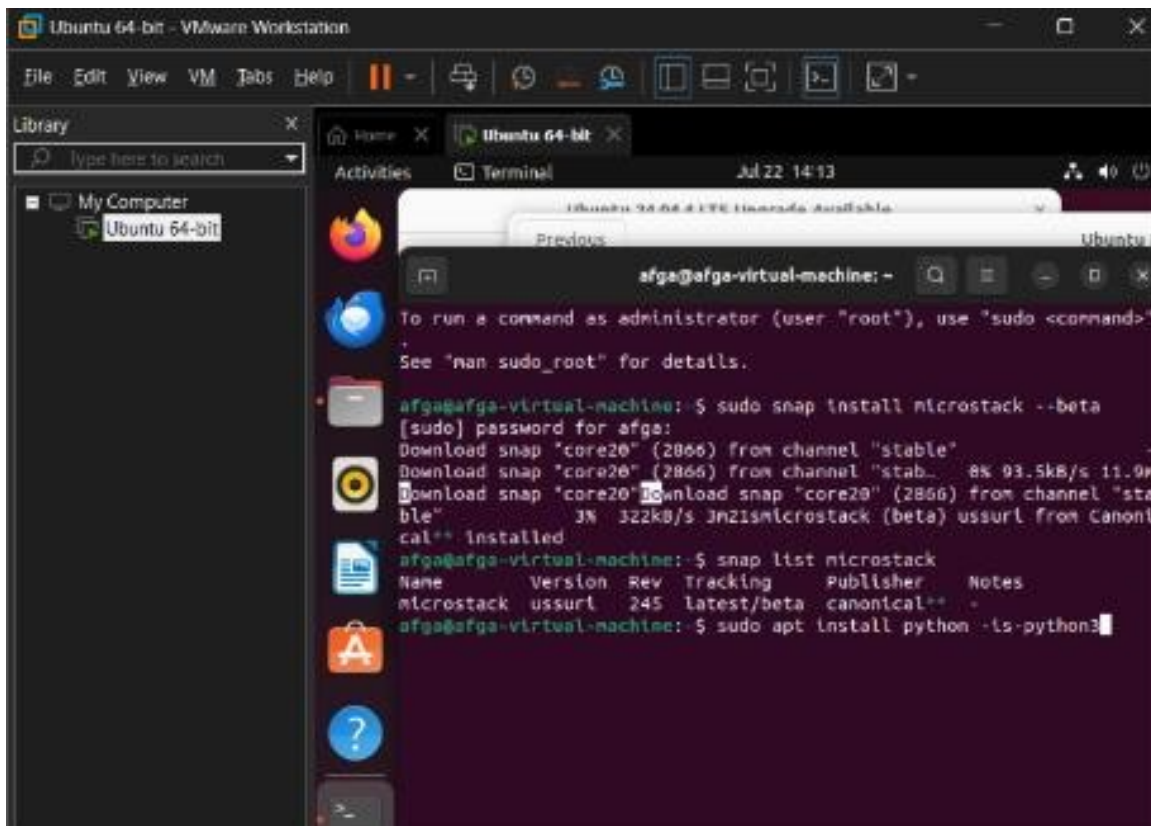
After the initialization process is completed, open a web browser inside the Ubuntu environment and enter the following address:

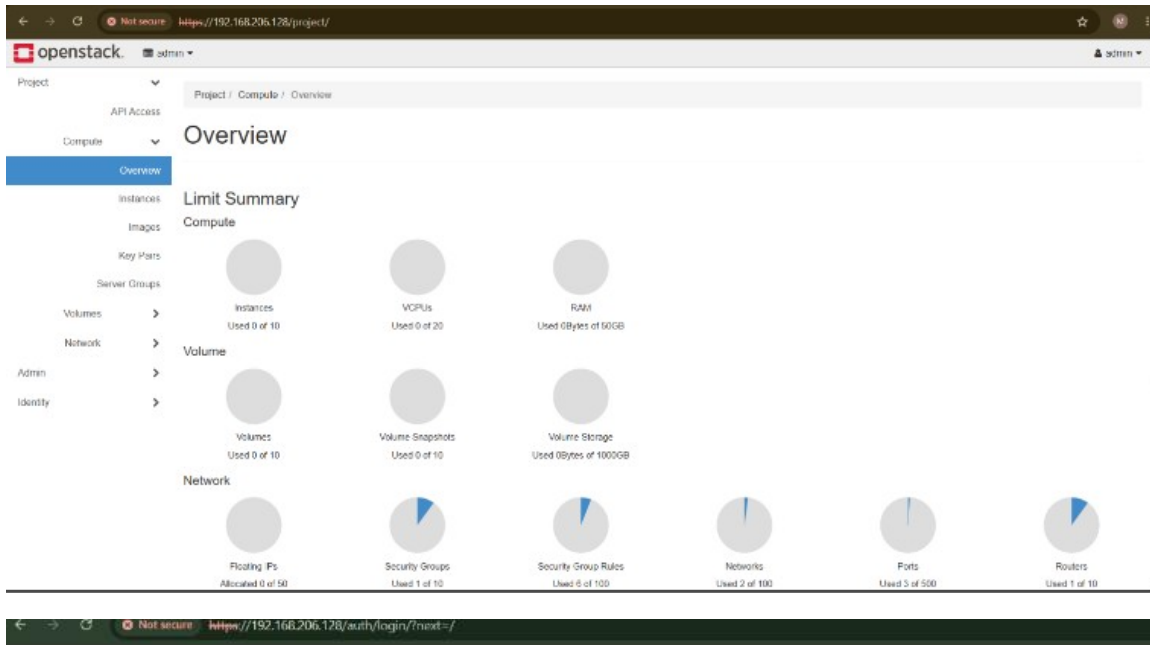
The OpenStack Horizon dashboard will then be displayed.

Use the following username to access the Horizon dashboard:

To obtain the automatically generated Keystone administrator password, execute the following command in the Ubuntu terminal:

The command displays the administrator password. Copy the displayed password and enter it along with the username admin on the Horizon login page.





The screenshot shows the OpenStack login page. It features the OpenStack logo at the top, followed by the text 'Log in'. Below this, there are two input fields: 'User Name' and 'Password'. The 'Password' field has a toggle icon for visibility. At the bottom right, there is a blue 'Sign In' button.

## Conclusion

A single-node Private Cloud environment was successfully set up using OpenStack (MicroStack). The OpenStack services were properly installed and configured, and the private cloud was able to launch, manage, and access virtual machine instances with functional compute, networking, and storage services, thereby meeting the objectives of the experiment.